Splitters: a neck above every pin

East bay 2: a test of funnel splitters · 13 November 2026

The Galton board's pins are fair but not independent: a ball leaving a pin still moves sideways and goes the same way again two times in three (`FINDINGS.md`, section 6). The board's first pin is the exception: fed straight down a narrow neck, it splits the balls 50/50. Could a neck above every pin make every decision a fresh one? If so, a board of such splitters would show the mathematics of independent fair coins exactly, and with different walls it could show more than the binomial (gambler's ruin, first passage, the reflection principle). This is the test, in bpp, before any bay is designed around it.

The test bed

`splitter-test.lua`, run on its own (not in the hall). Rows of "teeth": each a 4 mm pin on top of a solid wedge whose sides slope down (60 degrees) into vertical necks; each neck drops a ball onto the pin below, half a pitch to the side, its foot 8 mm above that pin's centre (as the board's neck). The same balls (6 mm steel), plates (a 7 mm gap) and contacts (bounce 0.4, friction 0.3) as the Galton board. Balls are fed one at a time down a neck onto the first pin, with a random offset of up to 0.3 mm; each ball's path is recorded in the board's format and measured with `paths.py`. Eight rows.

```
SP_PATHS=/tmp/sp.txt SP_BALLS=600 bpp -f splitter-test.lua -n 24000
python3 paths.py /tmp/sp.txt
```

Sizes and contacts are set from the environment (`SP_S` the pitch, `SP_NECK_W`, `SP_NECK_L`, `SP_SLOPE`, `SP_BOUNCE`, `SP_FRICTION`, ...). Over 70,000 balls went through the plain necks in all the runs below; none stopped.

The first try: the memory moved into the neck

At a 20 mm pitch with necks 8 mm wide and 12 mm long, the balls went the same way 72% of the time, worse than the lattice. Logging how balls leave a neck (`SP_LOG=1`) showed why: a ball crosses the neck, meets the far wall and stays nearer it, leaving 0.4 mm off centre on the side it was heading to (78% of the time). The next pin sees it off-centre on that side.

But each part of the shape pushes the memory one way or the other (300 balls each; P(same) is the chance of going the same way as at the last pin, 0.5 for fair coins):

at a 20 mm pitch	P(same)	correlation 1 row apart
necks 8 wide, 12 long (the first try)	0.72	+0.44
necks 6.6 wide	0.41	-0.19
necks 7 wide	0.26	-0.49
necks 10 wide	0.80	+0.61
necks 4 long	0.99	+0.98
necks 24 long	0.34	-0.32
sides 45 degrees	0.72	+0.43

at a 20 mm pitch	P(same)	correlation 1 row apart
sides 75 degrees	0.29	-0.41
bounce 0.7	0.43	-0.15
bounce 0.1	1.00	+1.00
friction 0.05	0.48	-0.05
pitch 30 mm	0.52	+0.05

Narrow or long necks and steep sides make a ball turn back; wide or short necks and a dead ball make it go on. (It looks as if the number of times a ball crosses the neck decides the sign, odd or even; not checked.)

A 30 mm pitch: as good as independent, at its design point

The longer slopes bring a ball to the neck faster, to cross it more. With 1,000 balls each:

at a 30 mm pitch	P(same)	correlation 1, 2, 3 rows apart	spread (8 fair coins: 1.414)
necks 8 wide, 12 long	0.53	+0.07, +0.01, -0.02	1.47
necks 10 long	**0.49**	**-0.01, 0.00, +0.01**	**1.43**
necks 15 long	0.51	+0.01, +0.01, +0.01	1.46
necks 7.5 wide	0.50	+0.01, +0.04, 0.00	1.46
necks 8.5 wide	0.54	+0.08, +0.01, 0.00	1.49
sides 55 degrees	0.42	-0.17, +0.06, -0.03	1.26
sides 65 degrees	0.40	-0.20, +0.04, +0.01	1.21
friction 0.2	0.55	+0.09, +0.02, -0.02	1.55
friction 0.45	0.43	-0.15, +0.01, -0.01	1.24
bounce 0.3	0.69	+0.39, +0.18, +0.07	2.05
bounce 0.5	0.42	-0.16, +0.01, +0.02	1.24
pitch 40 mm	0.38	-0.24, +0.04, -0.02	1.11

(The correlations' uncertainty is about 0.01.) With necks 10 mm long the decisions cannot be told from fair, independent coins: P(same) 0.494, no correlation at any distance, the spread 1.43 against 1.414. That is the default of the test bed now.

- **Where the memory is small, it is one pin deep.** Two and three rows apart the correlations are within 0.06 in all these, against -0.21 on the lattice: what a splitter remembers is which way the ball came in. (Where the memory is strong, as at a bounce of 0.3 with short necks, it reaches two rows: below.)
- **The shape is forgiving:** a millimetre on the necks' width or length keeps P(same) within 0.49 to 0.54.
- **The bounce is not:** 0.3 gives 0.69, 0.5 gives 0.42. The slope (55 or 65 degrees) and the friction matter about as much.

Longer, narrower necks: less sensitive, but taller

With more crossings in a neck the memory should average out. P(same), 600 balls each:

	bounce 0.25	0.4	0.55	row to row
pitch 30, necks 8 wide, 12 long	0.69 (at 0.3)	0.53	0.42 (at 0.5)	39 mm
pitch 30, necks 8 wide, 24 long	0.66	0.55	0.63	51 mm
pitch 30, necks 8 wide, 36 long	0.59	0.47	0.65	63 mm
pitch 30, necks 7 wide, 24 long	0.64	0.48	0.64	51 mm
pitch 30, necks 7 wide, 36 long	**0.42**	**0.53**	**0.44**	63 mm
pitch 40, necks 8 wide, 24 long	0.54	0.61	0.69	59 mm
pitch 24, necks 8 wide, 36 long	0.78	0.38	0.53	54 mm

Over bounces of 0.25 to 0.55, the best (necks 7 mm wide and 36 mm long) stays within 0.42 to 0.53, against 0.42 to 0.69 for the short necks. It helps, but does not take the bounce out, and twelve rows of it would stand 0.76 m tall (the board's pins take 0.13 m).

Taking the bounce out: three tries

Mixing chambers. Partway down the neck it widens around a small post and narrows again: a little coin flip whose result is thrown away, mirror-symmetric so it cannot favour a side. Short chambers (4 mm) trap balls between the post and the narrowing wall. Longer ones pass them but are worse at a low bounce: at 0.25 the balls turned back 86 to 90% of the time. Every mixer in a vertical neck changes how many times a ball crosses, which is what the bounce already does.

Leaning necks. Lean the neck, and gravity holds a ball on its lower wall: it rolls out the same way whichever side it came in from, the bouncing damped instead of counted. The pin then goes where the ball lands, and the lean changes side row by row, so what one row favours the next undoes. It erased too much: the balls left the neck so alike that the pin became a knife-edge again (1 mm of the pin's place took a row from 98% one way to 96% the other), and at the balance point the little that still varied was the part that remembered: memory +0.75. A splitter needs plenty of randomness beside what it remembers; erasing the memory is no use if the randomness goes with it. The vertical neck's crossings are what give it that randomness, and why the bounce matters.

A corner. Building the wedges from two convex slices instead of one puts a seam where the slope meets the neck, at the corner a ball rolls over into the neck. Nothing else changes, but at a 30 mm pitch and 10 mm necks the memory moved from -0.01 to +0.12 (1,000 balls each, both reproducible to the ball). How a ball meets that corner matters that much. A printed corner (its radius, the layer lines, wear) will not be what any simulation assumed, so no simulation can say a printed splitter's memory to better than about 0.15.

(The memory here is `paths.py`'s row-by-row measure: P(right | right at the last pin) - P(right | left at it), averaged over the rows. It is 0 for independent pins even where a pin favours a side, which a pooled correlation would mistake for memory.)

Tuning instead: the necks' length for the bounce

So the splitters have to be tuned on the board that is built: necks as inserts in two or three lengths, swapped until the balls stop favouring "the same way" or "back". As a starting point, the memory and the spread (8 rows; fair coins 1.414) at a 30 mm pitch with necks 8 mm wide, 800 balls each (the memory's uncertainty about 0.03):

bounce	necks 6	8	10	12	14	17	20 mm
0.3	+0.56, 2.46	+0.57, 2.47	+0.39, 2.05	+0.38, 2.04	+0.37, 2.06	+0.17, 1.58	**+0.06, 1.56**
0.4	+0.36, 1.97	**+0.03, 1.44**	**-0.02, 1.41**	**+0.07, 1.46**	-0.20, 1.20	-0.09, 1.36	-0.27, 1.14
0.5	-0.30, 1.09	-0.44, 0.95	-0.33, 1.06	-0.16, 1.26	-0.12, 1.32	**0.00, 1.39**	**-0.02, 1.39**
0.6	-0.25, 1.08	-0.36, 1.02	**-0.03, 1.36**	+0.06, 1.53	+0.09, 1.58	+0.24, 1.79	+0.27, 1.82

Every bounce tried has a length near nothing, but they do not follow in order (0.4 wants 10 mm, 0.5 wants 17, 0.6 wants 10 again), and with the seam at the corner the table moves by up to 0.15 (at 0.3 no length came under +0.4). Where the memory is strong it reaches two rows (at 0.3 with 6 mm necks, +0.37 two rows apart). The table says where to start; the board says where to stop.

What it means

- A neck above every pin can cut the memory to nothing, but only at a shape that suits the bounce, and how a ball meets the neck's mouth moves that shape. No passive mixer tried (chambers, a lean) took the bounce out.
- So a splitter board is to be tuned as built: necks as inserts in a few lengths (and the bounce measured first, `FINDINGS.md` section 5, to choose where to start from the table above). Watching a few hundred balls with `GB_PATHS`-style counting, or simply the spread of the pile against its engraved curve, says which way to go: too wide, the balls keep on (longer or shorter necks, by the table); too narrow, they turn back.
- Short necks keep the board's height reasonable (37 mm a row at a 30 mm pitch and 10 mm necks: twelve rows in 0.45 m; 20 mm necks, 0.57 m).
- Nothing jammed in the plain necks (over 70,000 balls in all), though the balls came one at a time; a full feed is still to try.